

Matthew Stuart Jones

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EDUCATION

University of Kentucky

Bachelor of Science, Computer Science

Projected Graduation, May 2027

GPA: 4.0/4.0

- Minors: Mathematics and Vocal Performance | Artificial Intelligence Certificate
- Study Abroad Experience in Bilbao, Spain (Universidad de Deusto)
- Dean's List — 6 Semesters

WORK EXPERIENCE

NSF Research Experience for Undergraduates (REU), AI for Social Good

Summer 2026

Worcester Polytechnic Institute (WPI) — Worcester, Massachusetts

- Developed ARISES, a novel feature selection method, outperforming all state-of-the-art baselines on 6 of 7 multi-label metrics
- Developed MLX, a counterfactual explainability method using structured gradient descent with label-co-occurrence-weighted gradients, outperforming SHAP and LIME on multi-label, patient-wise explanations for black-box classifiers
- Applied prompt engineering and AI tooling across the pipeline, matching tool to task for efficiency without losing accuracy

AI Researcher for ACCESS

October 2024 – Present (incl. Summer 2025)

UK Center for Computational Sciences (CCS) — Lexington, Kentucky

- Built MCP-based semantic router to categorize queries and route them to appropriate AI agents across the NSF ACCESS cyberinfrastructure ecosystem
- Developed Retrieval-Augmented Generation (RAG) system to help users draft allocation abstracts and proposals based on a corpus of past submissions
- Designed multi-LLM tournament tagger to categorize research proposals by field of science using open-source models deployed on a Grace Hopper cluster via Ollama
- Improved existing AI models through document verification and retraining workflows

Summer Teacher Intern

Summer 2024

Lavner Education — British International School Chicago, Chicago, Illinois

- Taught STEM-based courses to students ages 7–14, covering introductory programming and technology concepts

RESEARCH

Independent Study: Responsible AI Citation & Policy Influence Mapping

Spring 2026

Advisor: Dr. Brent Harrison, University of Kentucky — Human Centered AI, AI Ethics

- Analyzed citation flows between RAI literature (AIES, NeurIPS) and 18 national AI governance frameworks; identified three-way misalignment between paper production (fairness), citation impact (explainability), and policy language (accountability)
- Built end-to-end NLP pipeline using Semantic Scholar API, intfloat/e5-large-v2 embeddings, and cosine similarity scoring across a 20,000+ paper corpus
- Presented at the 20th Annual University of Kentucky Showcase of Undergraduate Scholars

SKILLS

- **AI/ML Engineering:** RAG systems, MCP semantic routers, LLM-based tagging pipelines; LlamaIndex, ChromaDB, sentence-transformers, Ollama; OpenAI API and local open-source models.
- **Programming & Systems:** Python (NumPy, pandas, scikit-learn); C and C++; SQL; JavaScript, HTML, CSS; Linux; R.
- **Research & Data Pipelines:** End-to-end NLP pipelines — API-based corpus ingestion, embedding generation, cosine similarity scoring; Semantic Scholar, OpenAlex APIs.
- **Policy Communication:** Translates AI/technical concepts for policy and general audiences; analyzed governance frameworks across multiple jurisdictions; fluent in Spanish.

ORGANIZATIONS AND INVOLVEMENT

Kentucky Alpha Chapter of Tau Beta Pi

Tau Beta Pi Scholar & Chapter Treasurer

- Managed chapter finances for a 120-member engineering honor society, including dues, budgeting, and scholarship requests
- Coordinated food and logistics for chapter meetings and events
- Volunteer judge at Future Cities regional engineering competition

acoUstiKats A Cappella

Business Manager & Social Media Manager

- Led the group's public presence and built an automated booking pipeline syncing forms with Google Calendar
- Spearheaded the group's first ICCA campaign where I initiated the decision to compete and led competition research
- Created content across platforms, designed and produced group merchandise, and publicized campaign events

UK Men's Chorus

Baritone Section Leader

- Serve as section leader on the group's New York tour, coordinating logistics and welfare for the baritone section
- Part of a leadership team focusing on culture and performance where singers are focused while taking care of each other